

	• Connects a Davis Vantage Pro2 or Vantage Vue to a BACnet network • BACnet IP over RJ45 compatible • Various selectable metric and imperial unit conversions • Provide additional alarms on gateway with two digital outputs (Open Collector Outputs)
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Connections

Connection	Description
VIN	Input Power Positive: 12-36VDC
GND	Input Power Negative (Ground)
DO0	Open Collector Output 0
DO1	Open Collector Output 1
USB-C & Female DE9 (Port 0)	For Firmware Update Only (DE9 and USB-C are mirrored); USB-C can be used as Power
D+	Not Used
D-	Not Used
Male DE9	Connection to Davis Weather Station
Ethernet Port	BACnet IP Port

Table 1: Connections to KTA-401

LEDs	Description
RST – Reset Button	Push Button • RED – Push button is pressed and held
BIP – BACnet IP	BACnet Communication Status LED • OFF – BACnet Communication is not/fail to be established, check BACnet Setups and Wiring • GREEN – Indicates good communication • RED – Push button is pressed and held for more than 3s (release the button to active the WiFi network)
VP2 – Weather Station	Weather Station Communication Status LED • OFF – Weather Station Communication is not/fail to be established, check Serial Connection • GREEN – Indicates good communication • RED – Push button is pressed and held for more than 10s (release the button to active Factory Reset Process)
STA - Status	Status LED • Flash once during start up • Solid ON: Normal Operation • Flash every 1s: Webpage Configuration Mode Enabled

Table 2: KTA-401 LED Functions

Configuring BACnet IP & Unit Conversion

By default, the device is set as below:

Parameters	Default Value
Ethernet Mode	Static IP Mode
IP Address	192.168.1.101
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS	1.1.1.1
Pressure Unit	1 – mmHg
Temperature Unit	1 – °C
Wind Speed Unit	1 – m/s
Rain Unit	1 – mm

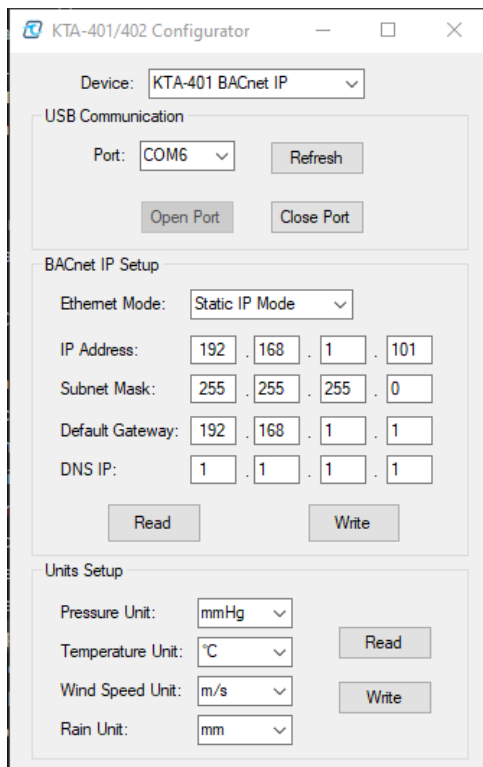
Table 3: Default Setups

There are four ways to adjust the BACnet IP and the units:

1. KTA-401/402 Configurator Software --- via USB C Port
2. Special ASCII Command --- via USB C Port
3. BACnet Write Property Function --- via RS485
4. Webpage --- via WiFi

1. KTA-401/402 Configurator Software

The baud rate for the USB-C port is set to **115200** and cannot be modify. To get started, simply select the appropriate COM port and open port. Once connected, you can easily read or write the BACnet IP and units setup using the corresponding buttons/text boxes.



2. Special ASCII Command

Like the configuration software, the Baud Rate for the USB-C port is fixed to be **115200** and cannot be modify.

	Command Sent	Command Received
Read BIP Setup	<i>R BIP\</i> <i>n</i>	<i>1\</i> <i>r\</i> <i>n</i>
Write BIP Setup	- Static Mode <i>W BIP Mode:0 IP:192.168.1.101 SN:255.255.255.0 GW:192.168.1.1 DNS:192.168.1.1\</i><i>n</i> - DHCP Mode <i>W BIP Mode:1\</i><i>n</i>	Write success – <i>OK\</i> <i>r\</i> <i>n</i> Write error – error message
Read Units	<i>R UNITS\</i> <i>n</i>	<i>P:1 T:1 S:1 R:1\</i> <i>r\</i> <i>n</i>
Write Units	<i>W UNITS P:0 T:1 S:2 R:0\</i> <i>n</i>	Write success – <i>OK\</i> <i>r\</i> <i>n</i> Write error – error message
Reboot the device	<i>REBOOT\</i> <i>n</i>	<i>"Rebooting..."</i>

Table 4: Special Command Table

Ethernet Mode: 0 – Static IP Mode, 1 – DHCP Mode

IP: Static IP Address

SN: Subnet Mask

GW: Default Gateway

DNS: DNS IP

For example, the command **"W BIP Mode:0 IP:192.168.1.101 SN:255.255.255.0 GW:192.168.1.1 DNS:192.168.1.1\n"** will set the device to Static IP mode with the IP address 192.168.1.101.

For the Read/Write Units command, the *P* is stand for *Pressure Unit*, *T* is *Temperature Unit*, *S* is *Wind Speed Unit*, and *R* is *Rain Unit*.

3. BACnet Write Property Function

The Ethernet Setup can be adjusted through the Proprietary-129 Object (see Table 14).

All changes will be saved and take effect once a 'TRUE' value is written to Proprietary Property-1002 and the system is power cycled.

Proprietary-129 Object	Description	Value
Proprietary Property – 990	BIP Ethernet Mode	0 – Static IP Mode 1 – DHCP Mode
Proprietary Property – 991	BIP IP Address	
Proprietary Property – 992	BIP Subnet Mask	
Proprietary Property – 993	BIP Default Gateway	
Proprietary Property – 994	BIP DNS IP	
Proprietary Property – 1002	Commit BACnet Setup to the memory	Write "TRUE" to save the BACnet setup to the memory ¹

Table 5: BACnet Setup Related Parameters

Note: After writing “TRUE” to the property 1002, you may receive a “Write Property Error” message on your BACnet Controller (BACnet Client device), due to the reinitialization of the Ethernet Connection.

Proprietary-129 Object	Description	Value
Proprietary Property - 1005	Pressure Unit	0 - inHg, 1 - mmHg, 2 - mbar
Proprietary Property - 1006	Temperature Unit	0 - °F, 1 - °C
Proprietary Property - 1007	Wind Speed Unit	0 - mph, 1 - m/s, 2 - fps, 3 - kph
Proprietary Property - 1008	Rain Unit	0 - inch, 1 - mm

Table 6: Unit Conversion Related Parameters

4. Webpage

The KTA-401 comes with a user-friendly web server, allowing users to easily modify the main settings via WiFi. Follow these steps to connect to the network:

- **Enable the KTA-401 WiFi network:** Press and hold the push button for 3s (until the BAC LED turns RED) then release it.
- **Activate network mode:** The status LED will start flashing every 1s, indicating that the network mode is active. This mode will remain active for 30 minutes.
- **Connect your device:** On your laptop or mobile, search for the WiFi network, which should appear as an SSID named “KTA401_xxxx” (where “xxxx” is the MAC address).
- **Join the network:** Connect using the default password: “password”
- **Access the webserver:** Open a web browser and enter the IP address “KTA401_xxxx.local” or “192.168.4.1”.
- The webserver’s configuration interface will open, allowing you to adjust the BACnet setup and providing a view current status of the KTA-401.

KTA-401 BACnet IP Weather Sta

192.168.4.1

Error



KTA-401 BACnet IP Weather Station Gateway

Real-time Data
Weatherstation Comm Status: **SUCCESS**
Barometer: 0.00 mmHg
Inside Temperature: -17.78°C
Outside Temperature: -17.78°C
Wind Speed: 4.47 m/s
Wind Direction: 10.00°
Instantaneous Rain Rate: 64.52 mm

RS232/USB (Port 0) Settings
Baud Rate: 115200

Ethernet Status
Ethernet Status: Static Mode
Ethernet IP: 192.168.1.101
Ethernet Subnet Mask: 255.255.255.0
Ethernet Gateway: 192.168.1.1
Ethernet DNS: 1.1.1.1

WiFi Status
WiFi Status: AP Mode

KTA-401 BACnet IP Weather Sta

192.168.4.1/configs

Error

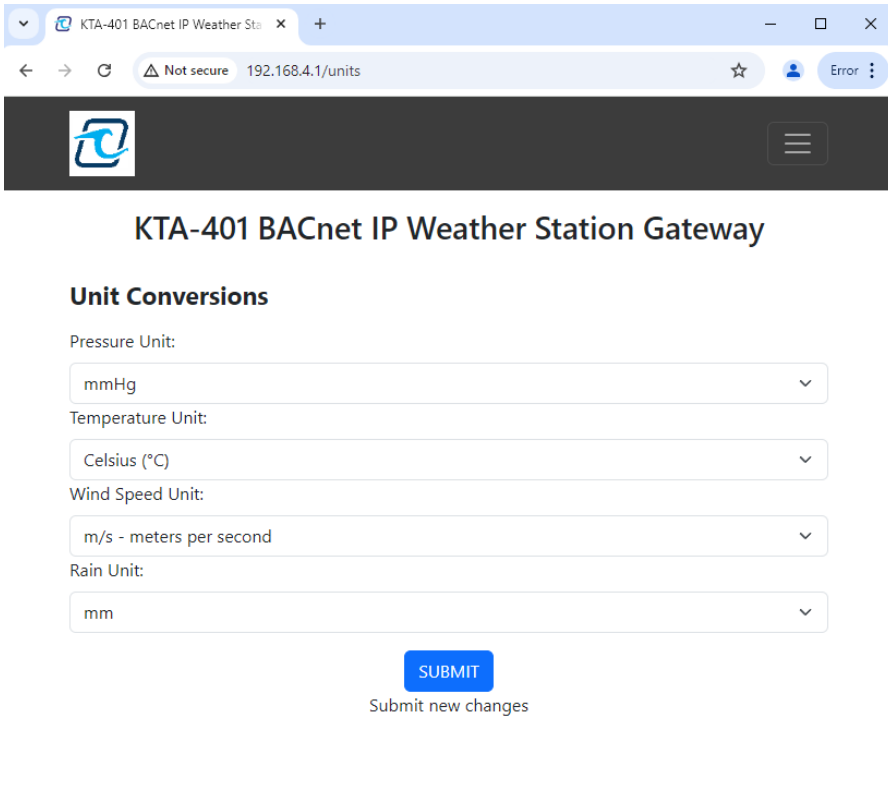


KTA-401 BACnet IP Weather Station Gateway

BACnet IP Settings
Ethernet Mode:
Static Mode
IP Address:
192.168.1.101
Subnet Mask:
255.255.255.0
Gateway:
192.168.1.1
DNS:
1.1.1.1

SUBMIT

Submit new changes



KTA-401 BACnet IP Weather Station Gateway

Unit Conversions

Pressure Unit:

mmHg

Temperature Unit:

Celsius (°C)

Wind Speed Unit:

m/s - meters per second

Rain Unit:

mm

SUBMIT

Submit new changes

Factory Reset:

The KTA-401 is equipped with a push button. To restore the factory settings for BACnet Communication, simply **press and hold the push button for 5s** while powered. This will set BACnet IP setup and Device ID back to the factory default.

Supported BACnet Services

- Read Property
- Read Property Multiple
- Write Property
- Device Communication Control
- Reinitialise Device
- Who-Has
- Who-Is

Supported BACnet Object Types

- Analog Input
- Analog Value
- Binary Value
- Device
- Data Value
- Integer Value
- Proprietary Value
- Time Value

Datalink Layer Option

- BACnet IP Server
- BACnet UDP Port: 47808 (0xBAC0)
- Static IP mode or DHCP mode
- BBMD function not implemented

Device Object

Property Name/ID	Access	Default
Object Identifier	R/W	838001
Object Name	R	KTA-401
Object Type	R	Device
System Status	R	Operational
Vendor Name	R	Ocean Controls
Vendor Identifier	R	838
Model Name	R	KTA-401 BACnet
Firmware Revision	R	V1.0
Software Version	R	1.3.6
Protocol Version	R	1
Protocol Revision	R	24
Max APDU Length	R	1476
Segmentation	R	None
APDU Timeout	R	3000
Number of APDU Retries	R	3

Table 7: BACnet Device Object Properties

Analog Input Objects

Analog Input Objects are the weather data read from the Davis Weather Station.

Property Name/ID	Access	Default	Note
Object Identifier	R		See table 7
Object Name	R		See table 7
Object Type	R	ANALOG INPUT	
Present Value	R		Value read from Weather Station
Status Flags	R	All false	*Alarm Flag set to TRUE when the Davis Weather Station alarm is reached.
Event State	R	NORMAL	
Out Of Service	R	FALSE	
Units	R		See table 9, 10 & 16

Table 8: BACnet Analog Input Object Properties

*The alarms need to be configured by WeatherLink Software.

Instance	Object Name	Default Unit	Supported Units
0	Barometer Trend 3h	No_Units	No_Units
1	Barometer	mmHg	inHg, mmHg, mbar
2	Inside Temperature	°C	°C, °F
3	Inside Humidity	%	%
4	Outside Temperature	°C	°C, °F
5	Wind Speed	m/s	mph, m/s, fps, kph
6	Wind Speed 10 Min	m/s	mph, m/s, fps, kph
7	Wind Direction	°	° (Degree Angular)
8	Extra Temp 1	°C	°C, °F
9	Extra Temp 2	°C	°C, °F
10	Extra Temp 3	°C	°C, °F
11	Extra Temp 4	°C	°C, °F
12	Extra Temp 5	°C	°C, °F

13	Extra Temp 6	°C	°C, °F
14	Extra Temp 7	°C	°C, °F
15	Soil Temp 1	°C	°C, °F
16	Soil Temp 2	°C	°C, °F
17	Soil Temp 3	°C	°C, °F
18	Soil Temp 4	°C	°C, °F
19	Leaf Temp 1	°C	°C, °F
20	Leaf Temp 2	°C	°C, °F
21	Leaf Temp 3	°C	°C, °F
22	Leaf Temp 4	°C	°C, °F
23	Outside Humidity	%	%
24	Extra Humidity 1	%	%
25	Extra Humidity 2	%	%
26	Extra Humidity 3	%	%
27	Extra Humidity 4	%	%
28	Extra Humidity 5	%	%
29	Extra Humidity 6	%	%
30	Extra Humidity 7	%	%
31	Rain Rate	mm	inch, mm
32	UV Index	No_Units	No_Units
33	Solar Radiation	W/m2	W/m2
34	Storm Rain	mm	in, mm
35	Day Rain	mm	in, mm
36	Month Rain	mm	in, mm
37	Year Rain	mm	in, mm
38	Day ET	mm	in, mm
39	Month ET	mm	in, mm
40	Year ET	mm	in, mm
41	Soil Moisture 1	millibar	millibar
42	Soil Moisture 2	millibar	millibar
43	Soil Moisture 3	millibar	millibar
44	Soil Moisture 4	millibar	millibar
45	Leaf Wetness 1	No_Units	No_Units
46	Leaf Wetness 2	No_Units	No_Units
47	Leaf Wetness 3	No_Units	No_Units
48	Leaf Wetness 4	No_Units	No_Units
49	Console Battery Voltage	V	V
50	Forecast Icon	No_Units	No_Units
51	Forecast Rule Number	No_Units	No_Units
52	Wet Bulb	°C	°C, °F
53	Wind Speed 2 Min	m/s	mph, m/s, fps, kph
54	Wind Gust	m/s	mph, m/s, fps, kph
55	Wind Gust Direction	°	° (Degree Angular)
56	Dew Point	°C	°C, °F
57	Heat Index	°C	°C, °F
58	Wind Chill	°C	°C, °F
59	THSW Index	°C	°C, °F
60	Rain Last 15 Min	mm	in, mm
61	Rain Last Hour	mm	in, mm
62	Rain Last 24 Hour	mm	in, mm
63	Barometric Calibration Number	mmHg	inHg, mmHg, mbar
64	Barometric Sensor Raw Reading	mmHg	inHg, mmHg, mbar

65	Absolute Barometric Pressure	mmHg	inHg, mmHg, mbar
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Table 9: Supported Analog Input Objects

Unit	Abbreviation	BACnet Code
Degrees Fahrenheit	°F	64
Degrees Celsius	°C	62
Percent Relative Humidity	%	29
Inches of Mercury	inHg	61
Millimetres of Mercury	mmHg	59
Millibar	mbar	134
Inches	in	32
Millimetres	mm	30
Miles per Hour	mph	78
Meters per Second	m/s	74
Feet per Second	fps	76
Kilometres per Hour	kph	75
Volts	V	5
Degrees Angular	°	90
Watts per Square Meter	W/m ²	35
No Units	No_Units	95

Table 10: Supported Analog Units

Time Value Objects

Time Value Objects are read from the Davis Weather Station.

Property Name/ID	Access	Default	Note
Object Identifier	R	0, 1	
Object Name	R	(0) Time of Sunrise (1) Time of Sunset	
Object Type	R	TIME (50)	
Present Value	R		Values are read from Weather Station
Status Flags	R	All false	

Table 11: BACnet Time Value Objects

These values require longitude and latitude to be set. Longitude and latitude can only be set by using the WeatherLink Software.

Date Value Object

Date Value Objects are read from the Davis Weather Station.

Property Name/ID	Access	Default	Note
Object Identifier	R	0	
Object Name	R	(0) Start Date of Current Storm	
Object Type	R	Date (42)	
Present Value	R		Value is read from Weather Station
Status Flags	R	All false	

Table 12: BACnet Date Value Objects

These values require longitude and latitude to be set. Longitude and latitude can only be set by using the WeatherLink Software.

Analog Value Objects

Analog Value Objects define the alarm setpoints. See Alarm Section for more information.

Property Name/ID	Access	Default	Note
Object Identifier	R	0, 1	
Object Name	R	(0) SP0 (1) SP1	
Object Type	R	ANALOG VALUE	
Present Value	R/W	0	
Status Flags	R	All false	
Event State	R	NORMAL	
Out Of Service	R	FALSE	
Units	R	No_Units	
Relinquish Default	R	0	
Proprietary Property - 3306	R/W	0	Alarm Operation Mode: - Data Type: Enumerated Alarm Active when: 0 - Equal Setpoint 1 - Greater than Setpoint 2 - Smaller than Setpoint 3 -Alarm Flag Active

Table 13: BACnet Analog Value Objects

Integer Value Objects

Integer Value Objects define which instance number (analog input object) will be controlled by alarm setpoints. See Alarm Section for more information.

Property Name/ID	Access	Default	Note
Object Identifier	R	0, 1	
Object Name	R	(0) Instance of SP0 (1) Instance of SP1	
Object Type	R	INTEGER VALUE	
Present Value	R/W	0	Instance number of Analog Input Object that should be controlled by alarm
Status Flags	R	All false	
Out Of Service	R/W	TRUE	Write TRUE to disable respective alarm
Units	R	No_Units	
Relinquish Default	R	0	

Table 14: BACnet Integer Value Objects

Binary Value Objects

Binary Value Objects show the status of the alarm outputs (digital outputs on the gateway). See Alarm Section for more information.

Property Name/ID	Access	Default	Note
Object Identifier	R	0, 1	
Object Name	R	(0) Digital Output 1 (1) Digital Output 2	
Object Type	R	BINARY VALUE	
Present Value	R	0	Status of Output: 0 – LOW, 0V 1 – HIGH, 3.3V
Status Flags	R	All false	
Event State	R	NORMAL	
Out Of Service	R	FALSE	
Polarity	R	NORMAL	

Table 15: BACnet Binary Value Objects

Proprietary Object - 129

The Proprietary Objects contains all the BACnet IP communication parameters and the status of the Weather Station Communication.

Property Name/ID	Access	Default	Note
Object Identifier	R	0	
Object Name	R	Configuration/Status	
Object Type	R	Proprietary Object - 129	
Proprietary Property - 990	R/W	FALSE	BIP Ethernet Mode - Data Type: Boolean - False: Static IP Mode - True: DHCP Mode
Proprietary Property - 991	R/W	192.168.1.101	BIP IP Address - Data Type: Octet String
Proprietary Property - 992	R/W	255.255.255.0	BIP Subnet Mask - Data Type: Octet String
Proprietary Property - 993	R/W	192.168.1.1	BIP Default Gateway - Data Type: Octet String
Proprietary Property - 994	R/W	192.168.1.1	BIP DNS IP Data Type: Octet String
Proprietary Property - 1002	R/W	FALSE	Commit BACnet Setup - Data Type: Boolean Write "TRUE" to save the BACnet setup to the memory
Proprietary Property - 1003	R	FALSE	Weather Station Loop 1 Status - Data Type: Boolean
Proprietary Property - 1004	R	FALSE	Weather Station Loop 2 Status - Data Type: Boolean
Proprietary Property - 1005	R/W	1	Pressure Unit - 0: inHg - 1: mmHg - 2: mbar

Proprietary Property - 1006	R/W	1	Pressure Unit - 0: °F - 1: °C
Proprietary Property - 1007	R/W	1	Wind Speed Unit - 0: mph - 1: m/s - 2: ft/s - 3: km/h
Proprietary Property - 1008	R/W	1	Rain Unit - 0: inch - 1: mm

Table 16: BACnet Proprietary Object

Alarms/Digital Outputs

The KTA-401 BACnet Gateway provides two configurable alarms which can enable or disable two digital outputs (Open Collector Outputs), see table 17 for the electrical parameters.

To enable the alarm function, the user needs to set FALSE to the OUT_OF_SERVICE property of the Integer Value Objects and configure the desirable parameters.

Alarm 0/Digital Output 0

Object Name	Object Name	Property	
Integer Value 0	Instance of SP0	Out Of Service	Enable or Disable the alarm
Integer Value 0	Instance of SP0	Present Value	Write the instance number of the analog input object to be controlled
Analog Value 0	SP1 (Setpoint 0)	Present Value	Write the desirable setpoint
Analog Value 0	SP1 (Setpoint 0)	Proprietary Property - 3306	Active the alarm when: 0 - Equal Setpoint 1 - Greater than Setpoint 2 - Smaller than Setpoint 3 -Alarm Flag Active
Binary Value 0	Digital Output 0	Present Value	Indicate the status of the DO1

Table 17: Alarm 1 Related Parameters

Alarm 1/Digital Output 1

Object Name	Object Name	Property	
Integer Value 1	Instance of SP1	Out Of Service	Enable or Disable the alarm
Integer Value 1	Instance of SP1	Present Value	Write the instance number of the analog input object to be controlled
Analog Value 1	SP1 (Setpoint 1)	Present Value	Write the desirable setpoint
Analog Value 1	SP1 (Setpoint 1)	Proprietary Property - 3306	Active the alarm when: 0 - Equal Setpoint 1 - Greater than Setpoint 2 - Smaller than Setpoint 3 -Alarm Flag Active
Binary Value 1	Digital Output 1	Present Value	Indicate the status of the DO2

Table 18: Alarm 2 Related Parameters

Symbol	Parameter	
Vmax	Maximum Allowable Voltage	36VDC
I_{max}	Maximum Allowable Current	200mA

Table 19: Digital Output Electrical Parameters

Additional Information**Device Communication Control (DCC)**

The DCC function enables or disables communication for devices on the BACnet network. When communication is disabled, the device will not respond to any requests except for RD or DCC requests. For security, this function is password protected - "**KTA-401_1234**".

Reinitialized Device (RD)

The RD function allows users to restart or reboot the gateway via BACnet.

For security, this function is password protected - "**KTA-401_1234**".